



欠驱动自主水下航行器 (AUV) 自适应固定时间反步三维轨迹跟踪控制

Adaptive fixed-time backstepping control for three-dimensional trajectory tracking of underactuated autonomous underwater vehicles

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ARTICLE INFO

Handling Editor: Prof. A.I. Incecik

Keywords:

Fixed time

3D trajectory tracking

Adaptive control

Backstepping

AUV

ABSTRACT

An adaptive fixed-time backstepping control is proposed to achieve the three-dimensional trajectory tracking control of an underactuated autonomous underwater vehicle (AUV) in the presence of model uncertainty and external disturbances. In this paper, the dynamics of the AUV in terms of five degrees of freedom (DOFs) are discussed. Considering it is an underactuated AUV, a virtual velocity guidance law is derived using the backstepping method. For velocity convergence, an adaptive fixed-time control is derived without model parameters, with adaptive adjusting law tackling system unknowns. Theoretical analyses demonstrate that the tracking error converges to a small bounded field within a fixed time in the proposed control scheme. The effectiveness and superiority of the proposed method are verified by simulation results.

1. Introduction

Over the past decades, autonomous underwater vehicles (AUVs) have attracted significant attention worldwide because of their capability to automatically perform underwater perception, monitoring, and operative missions (Shi et al., 2017; Huang et al., 2020). In order to accomplish the above missions, it is essential to investigate the trajectory tracking control of AUV. Since AUV is required to track the desired time-varying trajectory, rapid and precise tracking is crucial (Chen et al., 2022a). Most AUVs are underactuated. The trajectory tracking of underactuated AUVs is challenging as the hydrodynamic coefficients and time-varying external disturbances are unknown (Wang et al., 2019; Zhang et al., 2020).

Many researchers have attempted to overcome these difficulties, and several approaches have been proposed, including backstepping (Cho et al., 2019; Dong et al., 2016), sliding mode control (SMC) (Elmokadem et al., 2016; Chen et al., 2016), adaptive control (Li et al., 2019), neural network (NN) control (Zhang et al., 2020; Liu and Du, 2021) and fuzzy logic (Yu et al., 2018). All these controllers are aimed at two-dimensional trajectory tracking.

However, as the number of unactuated DOFs increases, it becomes difficult to control an AUV in a three-dimensional (3D) space (Wang et al., 2019). Consequently, only a few researchers have investigated the control problem. Through defining a spatial error, Li et al. derived a backstepping controller without decoupling, forcing the tracking error

to converge into a small field (Li et al., 2015). Motivated by (Li et al., 2015), Xu et al. first investigated the tracking problem of a control moment gyros (CMGs)-based AUV with input saturation (Xu et al., 2019). However, neither of the studies considered model uncertainty, and the spatial errors could not be eliminated, which decreased the tracking accuracy. System uncertainty is difficult to be characterized mathematically; therefore, the designed controller must overcome unexpected dynamics (Rodriguez et al., 2020). Xu et al. (2022) developed a radial basis function NN (RBFNN) and adaptive switching gain to cope with model uncertainty and external disturbances. Shojaei et al. derived a controller using input-output feedback linearization, with an NN approximator estimating the model uncertainty (Shojaei and Arefi, 2015). Yan et al. proposed an integral SMC for an underactuated AUV with an RBFNN that compensates systematic uncertain dynamics (Yan et al., 2019). Combining a multilayer NN and adaptive robust control, Shojaei et al. developed an NN-based feedback-linearizing controller, preventing undesirable transient state behavior and reducing the risk of actuator saturation (Shojaei, 2022). Although the methods mentioned above exhibit good performance, they only guarantee asymptotic stabilization; further, most of them compensate system unknowns by a NN, which increases complexity and computational cost.

In recent research, a novel adaptive SMC was proposed by Roy (Roy et al., 2020), which achieved uniformly ultimately boundedness without a priori bounded uncertainty of Euler-Lagrange system, based on which Li et al. developed an adaptive integral SMC and validated it in a surface

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<https://doi.org/10.1016/j.oceaneng.2023.114109>

Received 28 October 2022; Received in revised form 7 February 2023; Accepted 3 March 2023

Available online 15 March 2023

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vessel and unmanned aerial vehicle (Li et al., 2022). Without determining the underwater vehicle parameters, Zhu et al. combined adaptive SMC with tolerant control to achieve the trajectory tracking of an AUV (Zhu et al., 2021). Compared with NN-based methods, the adaptive SMC developed in these studies exhibits simplicity and low computational complexity. However, these methods only analyze the fully actuated system without considering the underactuated system. In addition, they only guarantee asymptotic convergence while neglecting the convergence rate.

Because quick response and precision are crucial for trajectory tracking (Chen et al., 2022a), a control method should be designed for rapid convergence and high accuracy. Thus, the finite-time stability has been proposed, which guarantees error convergence in the settling time. Yu et al. addressed global finite-time control for underactuated unmanned underwater vehicles based on proportional integral derivative-SMC, stabilizing all trajectory-tracking errors in finite time (Yu et al., 2019). Xia et al. investigated the 3D finite-time formation problem of a multi-underactuated AUV using a dual closed-loop fast integral terminal SMC, incorporating an RBFNN and a conditional integrator to address system nonlinearity (Xia et al., 2021). Liu et al. proposed a finite-time prescribed performance control for underactuated AUV based on an adaptive NN and tan-type BLF, realizing finite-time tracking convergence under the influence of output constraints and environmental disturbances (Liu et al., 2022). However, the settling time of finite-time control is determined by the initial errors. Fixed-time stability has been proposed to overcome this drawback. As an improvement of finite-time control, the upper boundary of its convergence time is made independent of initial errors. In (Chen et al., 2022b) and (Zheng et al., 2021), a fixed-time SMC was proposed for fully actuated underwater vehicles under time-varying external disturbances. In (Zhou et al., 2021; Zhang and Yang, 2020; Zhang and Wu, 2021), fixed-time trajectory tracking controllers for underactuated surface vessels were presented. An et al. investigated the 3D trajectory tracking control of underactuated AUV and realized fixed-time convergence, demonstrating that its convergence rate was faster than that of a finite-time controller (An et al., 2022). Although fixed-time control has been widely used in marine vehicles, most studies only investigated fully actuated systems or two-dimensional trajectory tracking. Few studies have investigated its application in the 3D trajectory tracking control of underactuated AUVs, which is a more challenging problem. Despite the achievements of An et al. in this area, their controller was derived based on model parameters and a precise dynamic model, which is difficult to apply practically.

Motivated by the research gap, we focused on adaptive fixed-time control for the 3D trajectory tracking of underactuated AUVs. For this purpose, a novel adaptive fixed-time backstepping control method is proposed. Considering it is an underactuated AUV, a virtual velocity guidance law is derived using the backstepping method to achieve 3D fixed-time trajectory tracking. Next, for the convergence of velocity, an adaptive adjusting law is proposed to tackle the model uncertainty and disturbances, because of which the fixed-time control of velocity is derived without model parameters. The stability and fixed-time convergence of the proposed control scheme are verified based on the Lyapunov theory, which guarantee that the tracking errors converge to a small bounded field within a fixed time and the spatial error is eliminated. Compared with existing adaptive methods (Xu et al., 2022; Shojaei and Arefi, 2015; Yan et al., 2019; Shojaei, 2022; Roy et al., 2020; Li et al., 2022; Zhu et al., 2021), the proposed method achieves fixed-time convergence, whose convergence time is independent of the initial errors. Compared with the existing fixed-time method (An et al., 2022), this controller is derived without model parameters and decoupling. In summary, the main contributions of this study are as follows.

- (1) A novel adaptive fixed-time controller is proposed for the 3D trajectory tracking of underactuated AUVs, which guarantees that the tracking error converges to a small bounded field in a fixed time.
- (2) Unlike the existing fixed-time controller, the proposed method is derived without model parameters, which is suitable for practical applications.
- (3) A novel adaptive adjusting law is proposed to tackle model uncertainty and external disturbances with a smooth and easily executable control input.

2. Mathematical model and preliminaries

In this section, the notations are defined, and the dynamics of the AUV are discussed in terms of five DOFs. Moreover, for the convenience of deriving a controller, relevant lemmas are introduced, and the aim of this study is defined.

2.1. Notations

For vector $\mathbf{I} = [I_1 \ I_2 \ \dots \ I_n]^T$ and matrix $\mathbf{A} \in \mathbf{R}^{n \times n}$, the norms are defined as $\|\mathbf{I}\| = \sum_j |I_j|$ and $\|\mathbf{A}\| = \sum_i \sum_j |A_{ij}|$, respectively. $I_j (j = 1, \dots, n)$ is the j th element of $\mathbf{I}$. A_{ij} is the element in the i th row and j th column of matrix $\mathbf{A}$. $\lambda_{\min}\{\mathbf{A}\}$ is the smallest eigenvalue of $\mathbf{A}$. $\text{sgn}(\cdot)$ and $\tanh(\cdot)$ denote the signum function and the hyperbolic tangent, respectively. $\text{sgn}(\mathbf{I})$ and $\tanh(\mathbf{I})$ are referred to as $\text{sgn}(\mathbf{I}) = [\text{sgn}(I_1) \ \text{sgn}(I_2) \ \dots \ \text{sgn}(I_n)]^T$ and $\tanh(\mathbf{I}) = [\tanh(I_1) \ \tanh(I_2) \ \dots \ \tanh(I_n)]^T$, respectively.

2.2. Mathematical model

According to (Yan et al., 2019), the kinematic and dynamic models of underactuated AUV in five DOFs can be defined as follows.

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta}) \mathbf{v} \quad \text{5 自由度, 二阶系统模型} \quad (1)$$

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{G}(\boldsymbol{\eta}) + \mathbf{f} = \boldsymbol{\tau} \quad (2)$$

where $\boldsymbol{\eta} = [\boldsymbol{\eta}_1; \boldsymbol{\eta}_2]^T$ is the vector of position and attitude in the earth-fixed frame, $\boldsymbol{\eta}_1 = [x \ y \ z]^T$, and $\boldsymbol{\eta}_2 = [\theta \ \psi]^T$. $\mathbf{v} = [\mathbf{v}_1; \mathbf{v}_2]^T$ is the vector of the linear and angular velocities expressed in the body-fixed frame, $\mathbf{v}_1 = [u \ v \ w]^T$, and $\mathbf{v}_2 = [q \ r]^T$. $\mathbf{J}(\boldsymbol{\eta}) \in \mathbf{R}^{5 \times 5}$ is the transformation matrix from the body-fixed frame to the earth-fixed frame, $\mathbf{M} \in \mathbf{R}^{5 \times 5}$ is the inertial matrix including added mass effects, $\mathbf{C}(\mathbf{v}) \in \mathbf{R}^{5 \times 5}$ is the matrix of centripetal and Coriolis forces, $\mathbf{D}(\mathbf{v}) \in \mathbf{R}^{5 \times 5}$ is the hydrodynamic damping matrix, $\mathbf{G}(\boldsymbol{\eta}) \in \mathbf{R}^5$ is a vector of gravitational and buoyancy forces, $\mathbf{f} \in \mathbf{R}^5$ represents the vector of external disturbances, and $\boldsymbol{\tau} \in \mathbf{R}^5$ is the vector of the control input. The detailed expressions are as follows.

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{O}_{3 \times 2} \\ \mathbf{O}_{2 \times 3} & \mathbf{J}_2 \end{bmatrix} \quad (3)$$

$$\mathbf{J}_1 = \begin{bmatrix} \cos \psi \cos \theta & -\sin \psi & \cos \psi \sin \theta \\ \sin \psi \cos \theta & \cos \psi & \sin \psi \sin \theta \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}, \mathbf{J}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1/\cos \theta \end{bmatrix} \quad (4)$$

$$\mathbf{M} = \text{diag}([M - X_{\ddot{u}} \ M - Y_{\ddot{v}} \ M - Z_{\ddot{w}} \ I_{yy} - M_{\ddot{q}} \ I_{zz} - N_{\ddot{r}}]) \quad (5)$$

$$\mathbf{C}(\mathbf{v}) = \begin{bmatrix} 0 & 0 & 0 & (M - Z_{\ddot{w}})w & -(M - Y_{\ddot{v}})v \\ 0 & 0 & 0 & 0 & (M - X_{\ddot{u}})u \\ 0 & 0 & 0 & -(M - X_{\ddot{u}})u & 0 \\ -(M - Z_{\ddot{w}})w & 0 & (M - X_{\ddot{u}})u & 0 & 0 \\ (M - Y_{\ddot{v}})v & -(M - X_{\ddot{u}})u & 0 & 0 & 0 \end{bmatrix} \quad (6)$$

$$\mathbf{D}(\mathbf{v}) = \text{diag}([X_{u|u}|u| + X_u Y_{v|v}|v| + Y_v Z_{w|w}|w| + Z_w M_{q|q}|q| + M_q N_{r|r}|r| + N_r]) \quad (7)$$

$$\mathbf{G}(\eta) = [0 \quad 0 \quad 0 \quad Z_B G \sin \theta \quad 0]^T \quad (8)$$

$$\boldsymbol{\tau} = [\tau_u \quad 0 \quad 0 \quad \tau_q \quad \tau_r]^T \quad (9)$$

The AUV system is divided into actuated and unactuated subsystems to simplify the controller design, and the former is expressed as follows (Shojaei and Arefi, 2015).

$$\mathbf{M}_a \mathbf{v} + \mathbf{C}_a \mathbf{v} + \mathbf{D}_a \mathbf{v} + \mathbf{G}_a + \mathbf{f}_a = \boldsymbol{\tau}_a \quad (10)$$

where $\mathbf{v} = [u \quad q \quad r]^T$, and detailed expressions of $\mathbf{M}_a$, $\mathbf{C}_a$, $\mathbf{D}_a$, $\mathbf{G}_a$, and $\boldsymbol{\tau}_a$ are as follows.

$$\mathbf{M}_a = \text{diag}([M - X_{\dot{u}} \quad I_{yy} - M_{\dot{q}} \quad I_{zz} - N_{\dot{r}}]) \quad (11)$$

$$\mathbf{C}_a = \begin{bmatrix} 0 & (M - Z_{\dot{w}})w & -(M - Y_{\dot{v}})v \\ (Z_{\dot{w}} - X_{\dot{u}})w & 0 & 0 \\ (X_{\dot{u}} - Y_{\dot{v}})v & 0 & 0 \end{bmatrix} \quad (12)$$

$$\mathbf{D}_a = \text{diag}([X_{u|u}|u| + X_u \quad M_{q|q}|q| + M_q \quad N_{r|r}|r| + N_r]) \quad (13)$$

$$\mathbf{G}_a = [0 \quad Z_B G \sin \theta \quad 0]^T \quad (14)$$

$$\boldsymbol{\tau}_a = [\tau_u \quad \tau_q \quad \tau_r]^T \quad (15)$$

According to the definitions and expressions above, the under-actuated AUV system has the following properties.

2.1.1. Property 1

The time derivative of the transformation matrix, $\mathbf{J}_1$, can be calculated by the following:

$$\dot{\mathbf{J}}_1 = \mathbf{J}_1 \mathbf{v}_2^* \quad (16)$$

where $\mathbf{v}_2^* = \begin{bmatrix} 0 & -r & q \\ r & 0 & 0 \\ -q & 0 & 0 \end{bmatrix}$ is a skew-symmetric matrix. For a vector $\mathbf{I}$, $\mathbf{I}^T \mathbf{v}_2^* \mathbf{I} = 0$.

2.1.2. Property 2

With the definitions of the norm in Section 2.1, matrices $\mathbf{M}_a$, $\mathbf{C}_a$, $\mathbf{D}_a$, $\mathbf{G}_a$, and $\mathbf{f}_a$ satisfy the following inequalities:

$$m_1 \leq \|\mathbf{M}_a\| \leq m_2, \|\mathbf{C}_a\| \leq c\|\mathbf{v}\|, \|\mathbf{D}_a\| \leq d_1 + d_2\|\mathbf{v}\|, \|\mathbf{G}_a\| \leq g, \text{ and } \|\mathbf{f}_a\| \leq f \quad (17)$$

where m_1 , m_2 , c , d_1 , d_2 , g , and f are unknown positive scalars.

2.2. Lemmas

Before proceeding, for the convenience of controller design and stability analysis, the following lemmas are introduced, referenced in Section 3. **大家看论文需要多积累类似的引理**

Lemma 1. (Yang and Ye, 2018)

For $a > 0$, $x \geq 0$, and $y > 0$, the following inequality holds.

$$x^a(y-x) \leq \frac{1}{1+a}(y^{1+a} - x^{1+a}) \quad (18)$$

Lemma 2. For $h > 1$, $y > 0$, and $x > 0$, the following inequality holds.

$$x^{2h} - y^{2h} \leq -(x-y)^{2h} + 2x^{2h} \quad (19)$$

Proof. When $0 < x \leq y$, define $f(y) = (y-x)^{2h} + x^{2h} - y^{2h}$, and $\partial f / \partial y = 2h(y-x)^{2h-1} - 2hy^{2h-1} \leq 0$. $f(y)$ decreases monotonically with $f(y=x) = 0$, and $x^{2h} - y^{2h} \leq -(x-y)^{2h} + 2x^{2h}$. When $0 < y < x$, define $f(x) = (x-y)^{2h} - x^{2h} - y^{2h}$, and $\partial f / \partial x = 2h(x-y)^{2h-1} - 2hx^{2h-1} < 0$. $f(x)$ decreases monotonously with $f(x=y) < 0$, thus yielding $x^{2h} - y^{2h} \leq -(x-y)^{2h} + 2x^{2h}$. The proof is completed.

Lemma 3. (Chen et al., 2022b)

Consider the following nonlinear system:

$$\dot{\mathbf{y}}(t) = \mathbf{f}(\mathbf{y}(t)) \quad (20)$$

where $\mathbf{y} = [y_1 \quad y_2 \quad \dots \quad y_n]^T$, $\mathbf{f}(\mathbf{y}) : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is continuous on $\mathbf{R}^n$, and $\mathbf{f}(\mathbf{0}) = \mathbf{0}$. For any vector $\mathbf{y}$, if scalars $q_1 > 0$, $q_2 > 0$, $0 < a < 1$, $b > 1$, $0 < \delta < \infty$ exist, with continuous positive definite function $V(\mathbf{y}) : \mathbf{R}^n \rightarrow \mathbf{R}^n$ satisfying the following condition

$$V(\mathbf{y}) > 0 \quad (21)$$

$$\dot{V}(\mathbf{y}) \leq -q_1(V(\mathbf{y}))^a - q_2(V(\mathbf{y}))^b + \delta \quad (22)$$

then the system is practically fixed-time stable and will adjust to the following residual set:

$$\Xi = \left\{ \lim_{t \rightarrow T_{\max}} \mathbf{y} | V(\mathbf{y}) \leq \min \left\{ q_1^{-\frac{1}{a}} \left(\frac{\delta}{1-\kappa} \right)^{\frac{1}{a}}, q_2^{-\frac{1}{b}} \left(\frac{\delta}{1-\kappa} \right)^{\frac{1}{b}} \right\} \right\} \quad (23)$$

where $0 < \kappa < 1$, and $T_{\max}$ is bounded by Eq. (24).

$$T_{\max} \leq \frac{1}{q_1 \kappa (1-a)} + \frac{1}{q_2 \kappa (b-1)} \quad (24)$$

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Lemma 4. (Chen et al., 2022b)

Define variables $y_1, y_2, \dots, y_n$ and a ; thus, the following inequalities hold.

$$\sum_{i=1}^n |y_i|^a \geq \left(\sum_{i=1}^n |y_i| \right)^a, 0 < a < 1 \quad \text{常用} \quad (25)$$

$$\sum_{i=1}^n |y_i|^a \geq n^{1-a} \left(\sum_{i=1}^n |y_i| \right)^a, a > 1 \quad (26)$$

Lemma 5. (Zhou et al., 2021)

For $\chi > 0$, the following inequality holds.

$$|x| \leq x \tanh(\chi x) + \frac{\varepsilon}{\chi}, \varepsilon = 0.2785 \quad (27)$$

2.4. Objective

The objective of this study is to solve the 3D trajectory tracking of an underactuated AUV subjected to model uncertainty and external disturbances, holding the tracking error converges to a small bounded field with fixed-time stability. In detail, supposing η_{1d} is the smooth desired 3D trajectory, the tracking error can be defined as follows.

$$\boldsymbol{\eta}_e = \boldsymbol{\eta}_1 - \boldsymbol{\eta}_{1d} \quad (28)$$

Without model parameters, derive out an adaptive fixed-time controller, such that the trajectory tracking error $\boldsymbol{\eta}_e$ satisfies the following condition:

$$\lim_{t \rightarrow T_{\max}} |\eta_{ej}| \leq o, (j=1, 2, 3) \quad (29)$$

where $T_{\max}$ is a constant representing the settling time, and o is a positive scalar.

核心思想, Backstepping

3. Adaptive fixed-time backstepping control

Motivated by (Li et al., 2015; Roy et al., 2020) and (Zhou et al., 2021), a novel adaptive fixed-time backstepping control is developed in this study to achieve the 3D trajectory tracking of underactuated AUVs. As shown in Fig. 1, the proposed control scheme consists of kinematic and dynamic control. Considering that it is an underactuated AUV, a virtual velocity guidance law is derived using the backstepping method in which the trajectory tracking error converges within a fixed time. Next, because it is difficult to obtain model parameters and disturbances, an adaptive velocity control law is proposed, guaranteeing the fixed-time convergence of the velocity.

3.1. Virtual velocity guidance law

Let η_{1d} represents the smooth desired trajectory. The virtual guidance point trajectory η_g is defined as follows.

$$\eta_g = \eta_{1d} - J_1 \rho \quad (30)$$

where $\rho = [\rho \ 0 \ 0]^T$ and $\rho > 0$ aid controller design. The error related to the virtual point can then be defined as follows.

$$\eta_{ge} = \eta_1 - \eta_g \quad (31)$$

The error, η_{ge} , should be expressed in a body-fixed frame to conveniently derive the controller:

$$e_g = J_1^T \eta_{ge} \quad (32)$$

$$\text{Define } z_1 = e_g - \rho \quad (33)$$

which satisfies $z_1 = J_1^T \eta_{ge} - \rho = J_1^T (\eta_1 - J_1 \rho - \eta_{1d}) - \rho = J_1^T \eta_e$. Therefore, z_1 is the trajectory tracking error expressed in a body-fixed frame. If $z_1 = 0$, then $\eta_e = 0$. Thus, the problem is transformed into how to stabilize z_1 . Based on (16) and (30)–(33), the derivative of z_1 is

$$-z_1^T K_2 \text{diag}(|z_1|^{2\beta-1}) \text{sgn}(z_1) \leq -k_2 z_1^T \text{diag}(|z_1|^{2\beta-1}) \text{sgn}(z_1) \leq -2^\beta 3^{1-\beta} k_2 \left(\frac{1}{2} z_1^T z_1 \right)^\beta \leq -2^\beta 3^{1-\beta} k_2 V_1^\beta \quad (41)$$

calculated as follows.

$$\begin{aligned} \dot{z}_1 &= -v_2^* e_g + J_1^T \dot{\eta}_{ge} - \dot{\rho} \\ &= -v_2^* z_1 - v_2^* \rho + v_1 - J_1^T \dot{\eta}_g - \dot{\rho} \\ &= -v_2^* z_1 + P v + [0 \ v \ w]^T - J_1^T \dot{\eta}_g - \dot{\rho} \end{aligned} \quad (34)$$

where $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -\rho \\ 0 & \rho & 0 \end{bmatrix}$, and $v = [u \ q \ r]^T$. The virtual velocity guidance law and the velocity error are defined as $v_d = [u_d \ q_d \ r_d]^T$ and $v_e = v - v_d$, respectively. Next, Eq. (34) is formulated as follows.

$$\dot{z}_1 = -v_2^* z_1 + P v_e + P v_d + [0 \ v \ w]^T - J_1^T \dot{\eta}_g - \dot{\rho} \quad (35)$$

For the fixed-time convergence of z_1 , the virtual velocity control law is derived as follows.

$$v_d = P^{-1} \left(-K_1 \tanh(z_1) - K_2 \text{diag}(|z_1|^{2\beta-1}) \text{sgn}(z_1) - [0 \ v \ w]^T + J_1^T \dot{\eta}_g + \dot{\rho} \right) \quad (36)$$

where K_1 and K_2 are the positive definite matrix, $1 < \beta < 2$.

Theorem 1. Considering the system in (35), if the virtual velocity control law is designed as (36) and $v_e = 0$, z_1 converges to a small bounded field in a fixed time.

Proof. The Lyapunov function candidate is selected as follows.

$$V_1 = \frac{1}{2} z_1^T z_1 \quad (37)$$

Based on (35), taking the time derivative of V_1 yields Eq. (38).

$$\dot{V}_1 = z_1^T (-v_2^* z_1 + P v_e + P v_d + [0 \ v \ w]^T - J_1^T \dot{\eta}_g - \dot{\rho}) \quad (38)$$

Considering that $z_1^T v_2^* z_1 = 0$, if $v_e = 0$, with the virtual velocity guidance law (36), (38) can be simplified as follows.

$$\dot{V}_1 = z_1^T \left(-K_1 \tanh(z_1) - K_2 \text{diag}(|z_1|^{2\beta-1}) \text{sgn}(z_1) \right) \quad (39)$$

Based on Lemmas 4 and 5, we obtain the following inequity:

$$-z_1^T K_1 \tanh(z_1) \leq k_1 (-z_1^T \text{sgn}(z_1) + 3\epsilon) \leq k_1 (-\sqrt{2} V_1^{0.5} + 3\epsilon) \quad (40)$$

where $k_1 = \lambda_{\min}(K_1)$, and $k_2 = \lambda_{\min}(K_2)$. Applying (40) and (41) to (39),

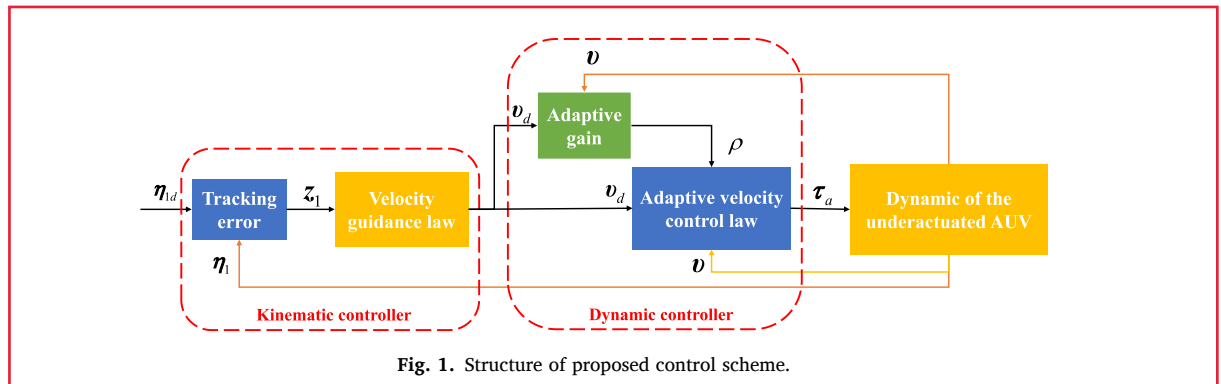


Fig. 1. Structure of proposed control scheme.

we obtain (42):

$$\dot{V}_1 \leq -aV_1^{0.5} - bV_1^\beta + 3k_1\varepsilon \quad (42)$$

where $a = \sqrt{2}k_1$, and $b = 2^\beta 3^{1-\beta}k_2$. According to Lemma 3, it can be

$$\begin{aligned} \dot{V}_3 &\leq -aV_1^{0.5} - bV_1^\beta + 3k_1\varepsilon + z_2^T \left(-\mathbf{K}_1 \tanh(z_2) - \mathbf{K}_2 |z_2|^{2\beta-1} \text{sgn}(z_2) + \mathbf{F} - \sigma \tanh(z_2) \right) \\ &\leq -aV_1^{0.5} - bV_1^\beta + 3k_1\varepsilon + z_2^T \left(-\mathbf{K}_1 \tanh(z_2) - \mathbf{K}_2 |z_2|^{2\beta-1} \text{sgn}(z_2) \right) - \|z_2\| \sum_{i=0}^2 (\hat{l}_i - l_i) \|\mathbf{v}\|^i + 3\sigma\varepsilon \end{aligned} \quad (48)$$

deduced that z_1 converges to a small bounded field at a fixed time. The proof is completed.

3.2. Adaptive velocity control law

According to Theorem 1, z_1 is stabilized with fixed-time stability if $\mathbf{v}_e = 0$. To guarantee the convergence of the velocity error, $\mathbf{v}_e$, an adaptive control law is designed as follows.

$$\tau_a = m \left(-P^T z_1 - \sigma \tanh(z_2) - \mathbf{K}_1 \tanh(z_2) - \mathbf{K}_2 |z_2|^{2\beta-1} \text{sgn}(z_2) \right) \quad (43)$$

where $m > m_2$, $z_2 = \mathbf{v}_e$, $\sigma = \hat{l}_0 + \hat{l}_1 \|\mathbf{v}\| + \hat{l}_2 \|\mathbf{v}\|^2$ is adaptive gains, and $\hat{l}_i$ is updated using Eq. (44):

$$\dot{\hat{l}}_i = \gamma_i \|z_2\| \|\mathbf{v}\|^i - \gamma_i k_1 \text{sgn}(\hat{l}_i) - 2\beta \gamma_i^{1-\beta} k_2 \hat{l}_i^{2\beta-1}, i = 0, 1, 2 \quad (44)$$

where γ_0 , γ_1 , and γ_2 are positive scalars. According to (44), $\hat{l}_i > 0$ if the initial value is positive.

Theorem 2. Considering the model of the AUV in (1)–(15), if the controller is designed as (36), (43) and (44), then the tracking error z_1 , velocity $\mathbf{v}_e$, and $\hat{l}_i$ will converge in a fixed time, leading to $\lim_{t \rightarrow T_{\max}} |\eta_{ej}| \leq 0$.

Proof. Let $z_2 = \mathbf{v}_e$; then, the Lyapunov function candidate is expanded as follows.

$$V_3 = V_1 + V_2 = \frac{1}{2} z_1^T z_1 + \frac{1}{2m} z_2^T M_a z_2 \quad (45)$$

Based on (10) and (42), taking the time derivative of V_3 , we obtain

$$\dot{V}_3 \leq -aV_1^{0.5} - bV_1^\beta + 3k_1\varepsilon + z_1^T P z_2 + z_2^T \left(\frac{1}{m} \tau_a + \mathbf{F} \right). \quad (46)$$

where $\mathbf{F} = m^{-1}(-C_a \mathbf{v} - D_a \mathbf{v} - G_a - f_a - M_a \ddot{\mathbf{v}}_d)$. According to Prop-

and using the adaptive control law (43), (46) can be simplified as follows.

Similar to (40) and (41), based on Lemmas 4 and 5 and considering $m/m_2 > 1$, we have the following inequalities.

$$\begin{aligned} -z_2^T \mathbf{K}_1 \tanh(z_2) &\leq k_1 \left(-(z_2^T z_2)^{0.5} + 3\varepsilon \right) \\ &\leq k_1 \left(-\sqrt{2} \left(\frac{m}{2m_2} z_2^T M_a z_2 \right)^{0.5} + 3\varepsilon \right) \\ &\leq k_1 \left(-\sqrt{2} V_2^{0.5} + 3\varepsilon \right) \end{aligned} \quad (49)$$

$$\begin{aligned} -z_2^T \mathbf{K}_2 |z_2|^{2\beta-1} \text{sgn}(z_2) &\leq -2^\beta 3^{1-\beta} k_2 \left(\frac{1}{2} z_2^T z_2 \right)^\beta \leq -2^\beta 3^{1-\beta} k_2 \left(\frac{m}{2m_2} z_2^T M_a z_2 \right)^\beta \leq \\ &-2^\beta 3^{1-\beta} k_2 V_2^\beta \end{aligned} \quad (50)$$

Applying (49) and (50) to (48), we obtain (51).

$$\dot{V}_3 \leq -aV_1^{0.5} - bV_1^\beta - aV_2^{0.5} - bV_2^\beta + 6k_1\varepsilon + 3\sigma\varepsilon - \|z_2\| \sum_{i=0}^2 (\hat{l}_i - l_i) \|\mathbf{v}\|^i \quad (51)$$

If $\hat{l}_i$ converges to l_i , then V_3 will converge within a fixed time. Therefore, the Lyapunov function candidate is expanded as follows:

$$V_5 = V_3 + V_4 = \frac{1}{2} z_1^T z_1 + \frac{1}{2m} z_2^T M_a z_2 + \frac{1}{2} \sum_{i=0}^2 \frac{\tilde{l}_i^2}{\gamma_i} \quad (52)$$

where $\tilde{l}_i = \hat{l}_i - l_i$. Taking the time derivative of V_5 and yields (53).

$$\dot{V}_5 = \dot{V}_3 + \sum_{i=0}^2 \frac{1}{\gamma_i} (\hat{l}_i - l_i) \dot{\hat{l}}_i \quad (53)$$

Based on the adaptive adjusting law (44), incorporating (51) and (53), we obtain (54).

$$\begin{aligned} \dot{V}_5 &= \dot{V}_3 + \|z_2\| \sum_{i=0}^2 (\hat{l}_i - l_i) \|\mathbf{v}\|^i - k_1 \sum_{i=0}^2 \text{sgn}(\hat{l}_i) (\hat{l}_i - l_i) - 2\beta k_2 \sum_{i=0}^2 \gamma_i^{-\beta} \hat{l}_i^{2\beta-1} (\hat{l}_i - l_i) \\ &\leq -aV_1^{0.5} - bV_1^\beta - aV_2^{0.5} - bV_2^\beta - k_1 \sum_{i=0}^2 \text{sgn}(\hat{l}_i) (\hat{l}_i - l_i) - 2\beta k_2 \sum_{i=0}^2 \gamma_i^{-\beta} \hat{l}_i^{2\beta-1} (\hat{l}_i - l_i) + 6k_1\varepsilon + 3\sigma\varepsilon \end{aligned} \quad (54)$$

erty 2, $\mathbf{F}$ satisfies the following condition:

$$\|\mathbf{F}\| \leq m^{-1} (c \|\mathbf{v}\|^2 + d_1 \|\mathbf{v}\| + d_2 \|\mathbf{v}\|^2 + g + f + m_2 \|\dot{\mathbf{v}}_d\|) \leq l_0 + l_1 \|\mathbf{v}\| + l_2 \|\mathbf{v}\|^2 \quad (47)$$

where l_0 , l_1 , and l_2 are unknown positive constants. Based on Lemma 5

Based on Lemmas 1, 2, and 4, we obtain the following inequalities.

$$\begin{aligned}
-k_1 \sum_{i=0}^2 \text{sgn}(\hat{l}_i)(\hat{l}_i - l_i) &= -k_1 \sum_{i=0}^2 (\hat{l}_i - l_i) \leq -2^{0.5} k_1 \sum_{i=0}^2 (0.5(\hat{l}_i - l_i)^2)^{0.5} \leq \\
&-2^{0.5} k_1 V_4^{0.5}
\end{aligned} \quad (55)$$

$$\begin{aligned}
-2\beta k_2 \sum_{i=0}^2 \gamma_i^{-\beta} \hat{l}_i^{2\beta-1} (\hat{l}_i - l_i) &\leq k_2 \sum_{i=0}^2 \gamma_i^{-\beta} (l_i^{2\beta} - \hat{l}_i^{2\beta}) \leq k_2 \sum_{i=0}^2 \gamma_i^{-\beta} (2l_i^{2\beta} - (\hat{l}_i - l_i)^{2\beta}) \\
&\leq k_2 \sum_{i=0}^2 \gamma_i^{-\beta} \left(2l_i^{2\beta} - 2^\beta \gamma_i^\beta \left(\frac{1}{2\gamma_i} (\hat{l}_i - l_i)^2 \right)^\beta \right) \leq -2^\beta 3^{1-\beta} k_2 V_4^\beta + k_2 \sum_{i=0}^2 2\gamma_i^{-\beta} l_i^{2\beta}
\end{aligned} \quad (56)$$

Based on Lemma 4 and incorporating (54)–(56), the following inequalities are derived:

$$\begin{aligned}
\dot{V}_5 &\leq -aV_1^{0.5} - bV_1^\beta - aV_2^{0.5} - bV_2^\beta - aV_4^{0.5} - bV_4^\beta + k_2 \sum_{i=0}^2 2\gamma_i^{-\beta} l_i^{2\beta} + 6k_1\epsilon + 3\sigma\epsilon \\
&\leq -aV_5^{0.5} - 3^{1-\beta} bV_5^\beta + \delta = -aV_5^{0.5} - b_1 V_5^\beta + \delta
\end{aligned} \quad (57)$$

where $b_1 = 3^{1-\beta}b$, and $\delta = k_2 \sum_{i=0}^2 2\gamma_i^{-\beta} l_i^{2\beta} + 6k_1\epsilon + 3\sigma\epsilon$. According to Lemma 3, with $0 < \kappa < 1$, V_5 regulates to the following residual set:

$$\Xi = \left\{ V_5 \leq \min \left\{ a^{-2} \left(\frac{\delta}{1-\kappa} \right)^2, b_1^{-\frac{1}{\beta}} \left(\frac{\delta}{1-\kappa} \right)^{\frac{1}{\beta}} \right\} \right\} \quad (58)$$

within settling time $T_{\max}$, which is bounded by the following inequality:

$$T_{\max} \leq \frac{2}{a\kappa} + \frac{1}{b_1\kappa(\beta-1)} \quad (59)$$

Therefore, z_1 , v_e , and $\tilde{l}_i$ converge to a small bounded field within a fixed time. In particular, because $z_1 = J_1^T \eta_e$, we derive Eq. (60).

$$\lim_{t \rightarrow T_{\max}} |\eta_{ej}| \leq o \quad (60)$$

where o is a small positive scalar determined by (58). The proof is

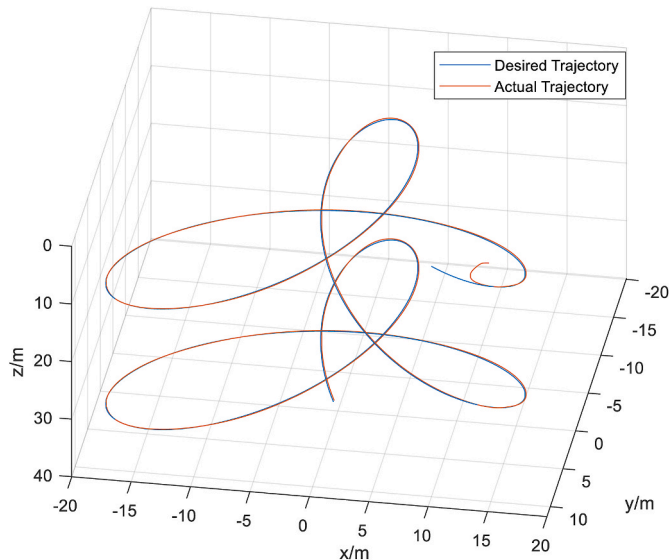


Fig. 2. Trajectory tracking in 3D space.

completed.

3.3. Main results

For the 3D trajectory tracking control of an underactuated AUV described by Eqs. 1–15, employ the virtual velocity guidance law:

$$\begin{aligned}
v_d &= P^{-1} \left(-K_1 \tanh(z_1) - K_2 \text{diag}(|z_1|^{2\beta-1}) \text{sgn}(z_1) - [0 \quad v \quad w]^T + J_1^T \dot{\eta}_g \right. \\
&\quad \left. + \dot{\rho} \right)
\end{aligned} \quad (61)$$

and the adaptive velocity control law:

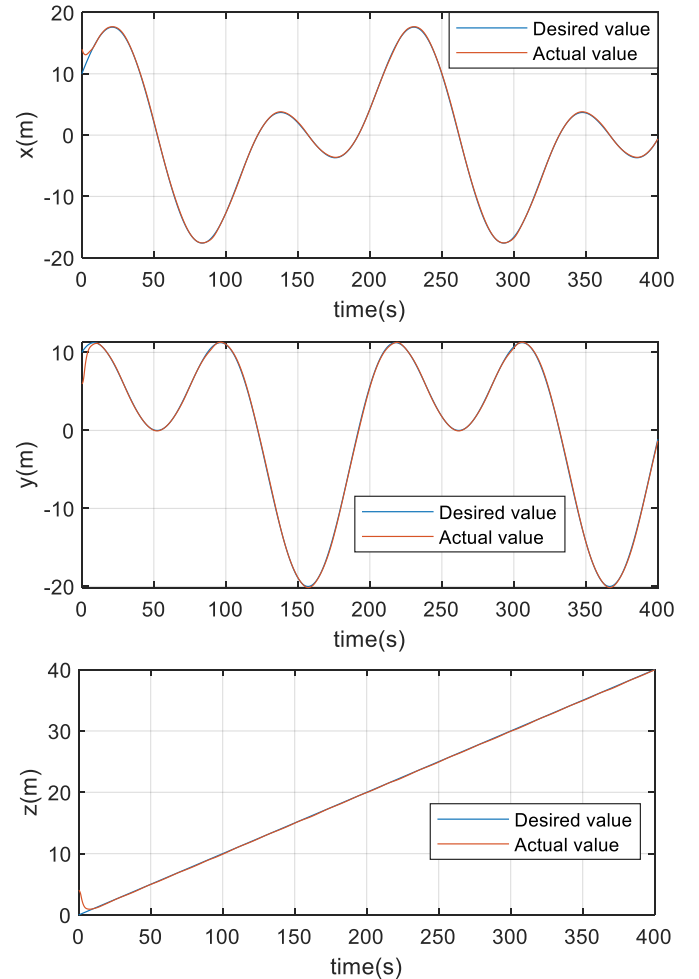


Fig. 3. Desired and actual values.

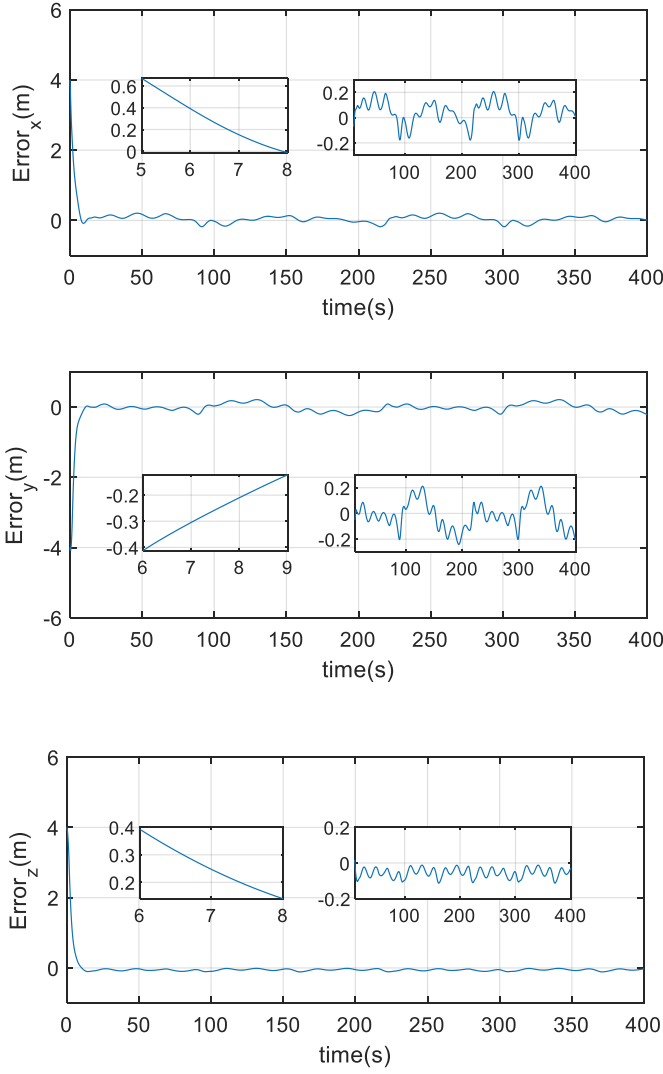


Fig. 4. Position errors.

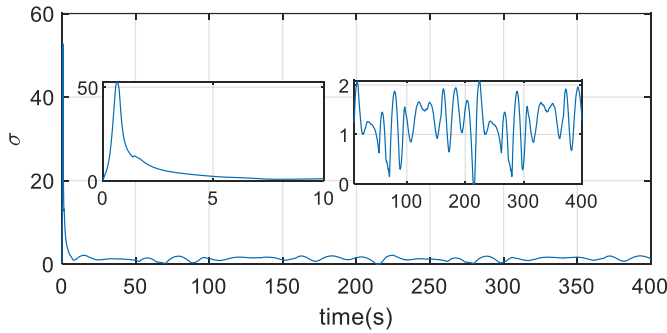


Fig. 5. Adaptive gains.

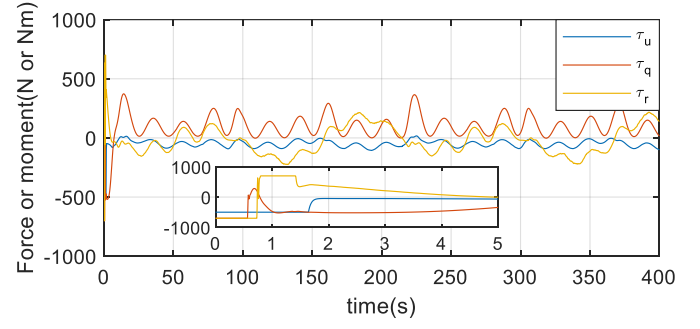


Fig. 6. Control forces and moments.

$$\begin{aligned} \tau_a &= m \left(-P^T z_1 - \sigma \tanh(z_2) - K_1 \tanh(z_2) - K_2 |z_2|^{2\beta-1} \text{sgn}(z_2) \right) \\ \sigma &= \hat{l}_0 + \hat{l}_1 \|v\| + \hat{l}_2 \|v\|^2, \hat{l}_i \\ &= \gamma_i \|z_2\| \|v\|^i - \gamma_i k_1 \text{sgn}(\hat{l}_i) - 2\beta \gamma_i^{1-\beta} k_2 \hat{l}_i^{2\beta-1}, i = 0, 1, 2 \end{aligned} \quad (62)$$

Then the trajectory tracking error, η_{ej} , converges to the small bounded field within settling time $T_{\max}$, that is, $\lim_{t \rightarrow T_{\max}} |\eta_{ej}| \leq o$. The proof is presented in Sections 3.1 and 3.2.

4. Numerical simulations

In this section, a trajectory tracking simulation and a comparison study are conducted to demonstrate the effectiveness and superiority of the proposed control scheme. The model parameters of the AUV are obtained from Ref (Li et al., 2015), which are as follows: $M = 183$, $I_{xx} = 3$, $I_{yy} = 95$, $I_{zz} = 95$, $X_{u_i} = -49$, $Y_v = -243$, $Z_w = -230$, $M_q = -140$, $N_r = -160$, $X_{u_i} = -13$, $Y_{\dot{v}} = -257$, $Z_{\dot{w}} = -257$, $N_{\dot{r}} = -86$,

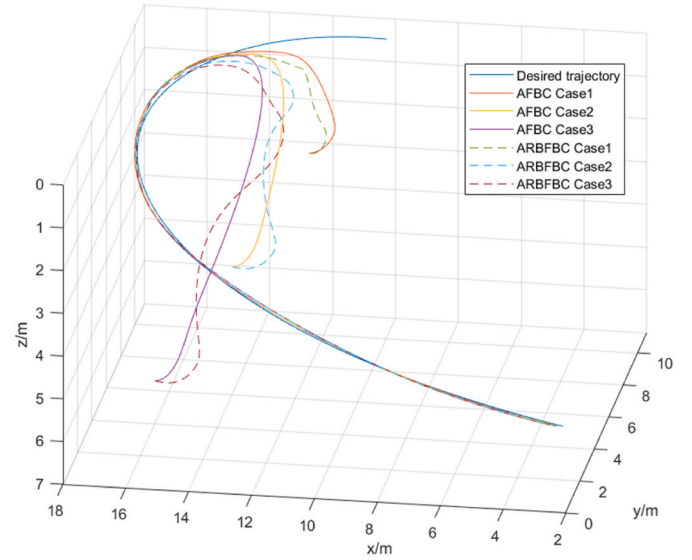


Fig. 7. 3D trajectory tracking results for two control schemes in three cases.

$M_{\dot{q}} = -86$, $X_{u|u|} = -16$, $Y_{v|v|} = -542$, $Z_{w|w|} = -422$, $M_{q|q|} = -62$, and $N_{r|r|} = -78$. In addition, the parameters of the proposed controller are as follows: $m = 300$, $\rho = e^{-3t} + 0.1$, $\beta = 13/11$, $\gamma_0 = 5$, $\gamma_1 = 0.5$, $\gamma_2 = 0.2$, $K_1 = \text{diag}([0.2 \ 0.2 \ 0.3])$, and $K_2 = \text{diag}([0.2 \ 0.2 \ 0.2])$. The maximum and minimum values of control inputs τ_u , τ_q , and τ_r are set as $\pm 500 \text{ N}$, $\pm 700 \text{ Nm}$, and $\pm 700 \text{ Nm}$, respectively. The ocean current is simulated by the following first-order Gauss-Markov process according to (Wang et al., 2015):

$$\dot{v}_c + \mu v_c = \varpi \quad (63)$$

where v_c is the magnitude of the ocean current, $\mu = 1$ and $\varpi = N(0.5, 0.3^2)$ represents Gaussian white noise. In this study, the direction of the ocean current is along the x -axis, and the external disturbances are set as

$$\tau_d = [30 \sin(0.3t + \pi/5) + 10, 30 \sin(0.3t + \pi/4) + 10, 15 \sin(0.3t + \pi/4)w + 10, 30 \sin(0.3t), 30 \sin(0.3t)]^T.$$

4.1. Trajectory tracking performance

Analysis of the trajectory tracking result is presented in this subsection. To verify the effectiveness of the proposed controller, a high maneuvering trajectory (Xu et al., 2022) and initial states are selected as $\eta_{1d} = [10 \sin(0.06t) + 10 \cos(0.03t) \ 10 \sin(0.03t) + 10 \cos(0.06t) \ 0.1t]^T$ and $\eta(0) = [14 \ 6 \ 4 \ 0 \ 0]^T$, respectively. The simulation results are presented in Figs. 2–6.

Fig. 2 shows the trajectory tracking results in 3D space, and Fig. 3 shows the time-varying curves of the desired and actual values. The results indicate that in the proposed control scheme, the underactuated

Table 1

Comparison of performance index.

Performance index	$\ e\ _{IAE}$	$\ \tau\ _{ECI}$	$T_{\max}$
AFBC Case1	1.03	376	7.4
ARBFBC Case1	1.23	399	8
AFBC Case2	2.30	430	8
ARBFBC Case2	2.54	590	10
AFBC Case3	3.72	484	8.5
ARBFBC Case3	4.07	641	10.7

AUV successfully tracks the desired 3D trajectory with model uncertainty and external disturbances. Fig. 4 shows further details of the position errors. All the trajectory tracking errors converge to $[-0.2, 0.2]$ within 8 s without overshooting and, subsequently, never exceed the values. The adaptive gains and control inputs are presented in Figs. 5 and 6 respectively. The adaptive gain increases to a high value in the early stage, owing to the large initial errors, then decreases rapidly, and converges at approximately 8 s. Consequently, the control inputs are adjusted to a high value, causing the trajectory tracking error to converge rapidly and change slowly in the stable region.

The simulations show that, in the presence of model uncertainty and external disturbances, although the proposed controller is derived without model parameters, it is still effective for 3D trajectory tracking of underactuated AUVs. Even if the initial position errors reach up to 4, -4 , and 4, the AUV spends 8 s in tracking the desired trajectory without any overshooting, maintaining the tracking error in a small bounded field. Furthermore, the control inputs are smooth without chattering.

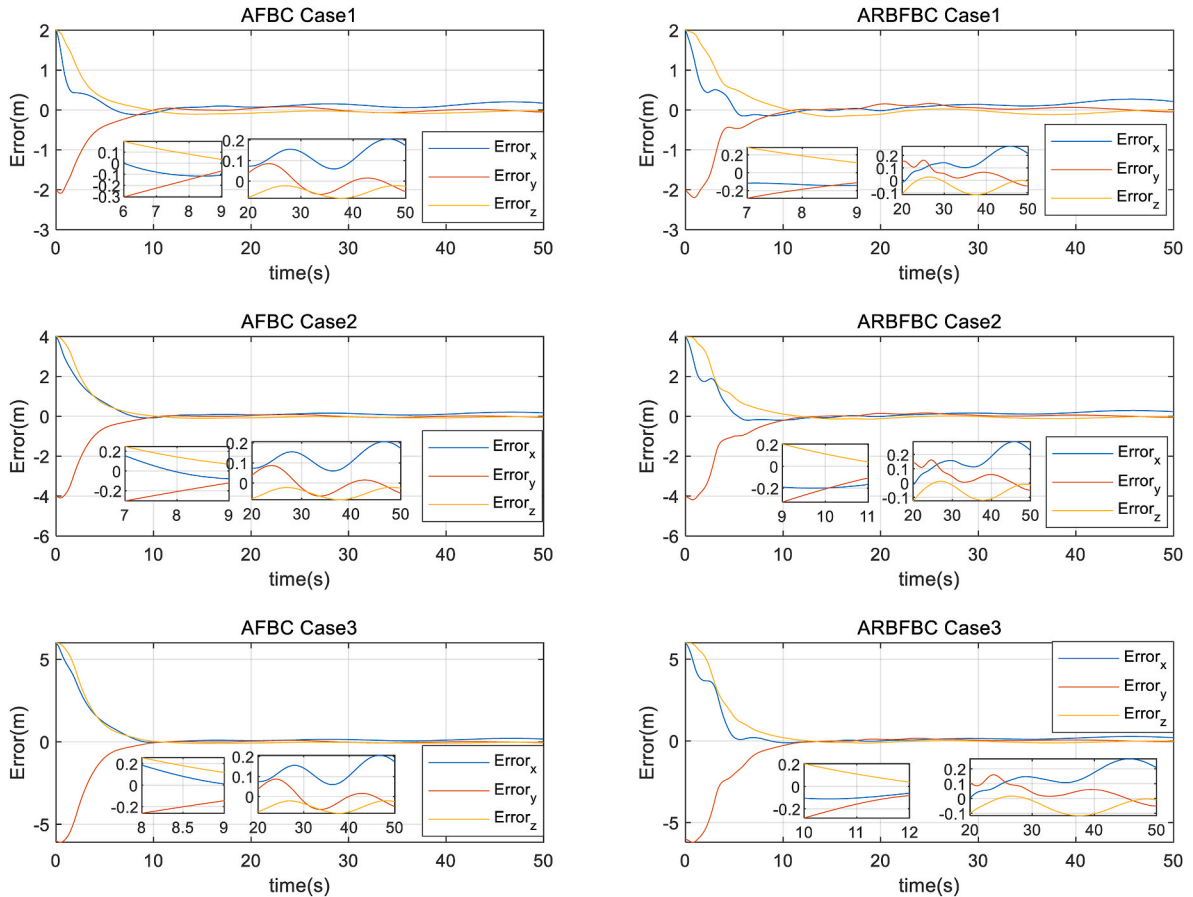


Fig. 8. Errors of two control schemes in three cases.

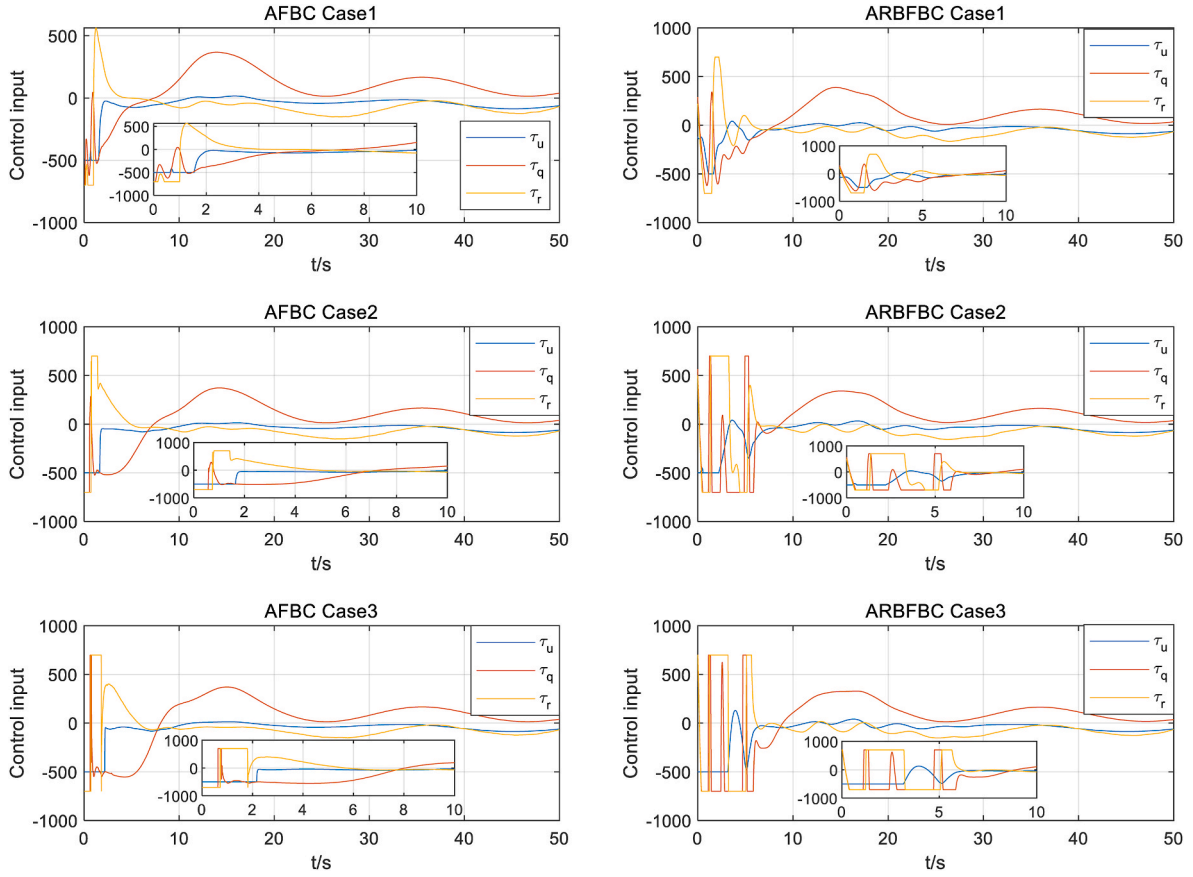


Fig. 9. Control input of two control schemes in three cases.

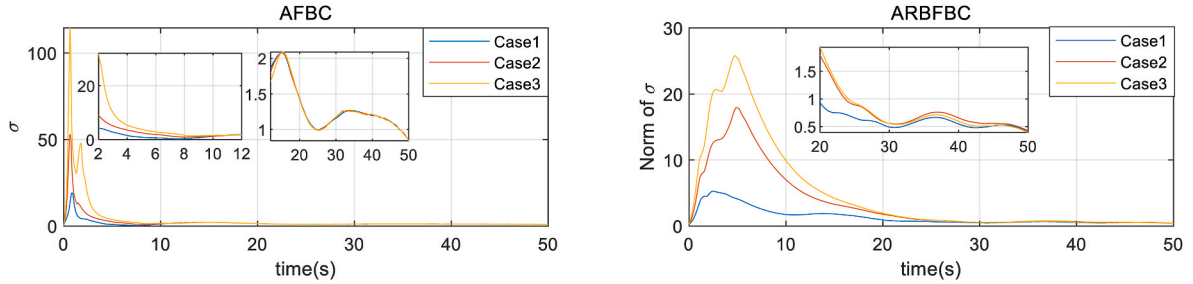


Fig. 10. Adaptive gain of two control scheme in three cases.

4.2. Comparison study

A comparison study of the adaptive RBF backstepping controller (ARBFBC) (Xu et al., 2022) is conducted in this subsection to demonstrate the superiority of the proposed adaptive fixed-time backstepping controller (AFBC). $e = J_1^T \eta_e$, $z_1 = e - \rho$, $[u_e \ q_e \ r_e]^T = v - v_d$, and $\sigma = [\sigma_u \ \sigma_q \ \sigma_r]^T$ are defined, and ARBFBC can be summarized as follows.

$$v_d = P^{-1}(-\Lambda_1 \tanh(z_1) - [0 \ v \ w]^T + J_1^T \dot{\eta}_d + \dot{\rho}) \quad (64)$$

$$\begin{aligned} \tau_u &= (M - X_u)(-l_u u_e - (e_1 - \rho) - \sigma_u \tanh(u_e) - W_u^T \Phi(v)) \\ \tau_q &= (I_{yy} - M_{\dot{q}})(-l_q q_e - \rho e_3 - \sigma_q \tanh(q_e) - W_q^T \Phi(v)) \\ \tau_r &= (I_{zz} - N_r)(-l_r r_e - \rho e_2 - \sigma_r \tanh(r_e) - W_r^T \Phi(v)) \end{aligned} \quad (65)$$

$$\begin{aligned} \dot{W}_u &= \Gamma_u u_e \Phi(v) - \gamma_u W_u \\ \dot{W}_q &= \Gamma_q q_e \Phi(v) - \gamma_q W_q \\ \dot{W}_r &= \Gamma_r r_e \Phi(v) - \gamma_r W_r \end{aligned} \quad (66)$$

$$\begin{aligned} \dot{\sigma}_u &= \lambda_u \tanh(u_e) - \mu_u \sigma_u \\ \dot{\sigma}_q &= \lambda_q \tanh(q_e) - \mu_q \sigma_q \\ \dot{\sigma}_r &= \lambda_r \tanh(r_e) - \mu_r \sigma_r \end{aligned} \quad (67)$$

These parameters are defined as $\Lambda_1 = \text{diag}([0.35 \ 0.35 \ 0.35])$, $\rho = [\rho \ 0 \ 0]^T$, $\rho = e^{-3t} + 0.1$, $l_u = 0.8$, $l_q = 0.8$, $l_r = 0.8$, $\Gamma_u = 1$, $\Gamma_q = 2$, $\Gamma_r = 1$, $\gamma_u = 0.1$, $\gamma_q = 0.1$, $\gamma_r = 0.1$, $\lambda_u = 1$, $\lambda_q = 1$, $\lambda_r = 0.5$, $\mu_u = 0.2$, $\mu_q = 0.2$, and $\mu_r = 0.2$. To further quantize the performance, the following integrated absolute error (IAE) and energy of control input (ECI) (Zhang et al., 2019) are introduced:

$$\begin{aligned} \|e\|_{IAE} &= \sqrt{\frac{1}{N} \sum_{k=1}^N \|e(k)\|^2} \\ \|\tau\|_{ECI} &= \sqrt{\frac{1}{N} \sum_{k=1}^N \|\tau(k)\|^2} \end{aligned} \quad (68)$$

In addition, the parameters of the AFBC, desired trajectory, and external disturbances are set to be the same as those described in Section 4.1, and the initial states are selected as in the following three cases.

$$\begin{aligned} 1 \quad \eta(0) &= [12 \quad 8 \quad 2 \quad 0 \quad 0]^T \\ 2 \quad \eta(0) &= [14 \quad 6 \quad 4 \quad 0 \quad 0]^T \\ 3 \quad \eta(0) &= [16 \quad 4 \quad 6 \quad 0 \quad 0]^T \end{aligned}$$

The comparative simulation results are presented as follows.

Fig. 7 shows the 3D trajectory tracking results of the two control schemes in three cases. The trajectories of AFBC in the three cases and ARBFBC in Case 1 converge to one point, whereas that of ARBFBC in Cases 2 and 3 are not. According to Fig. 8 and Table 1, the convergence times of AFBC in the three cases are 7.4, 8, and 8.5 s, whereas those of ARBFBC are 8, 10, and 10.7 s, which indicates that as the initial error increases AFBC exhibits a faster convergence rate than ARBFBC. In addition, the error vibration and IAE of ARBFBC is larger than that of AFBC. Figs. 9 and 10 show the control input and adaptive gain, respectively. In the three cases, the control inputs of the AFBC are smooth and easily executable, and the adaptive gain increases significantly and then converges within 8 s, while the control input of ARBFBC exhibits severe oscillation and larger ECI in Cases 2 and 3. Additionally, the convergence time of the adaptive gain increases with increasing initial error. The gain of the ARBFBC is smaller than that of the AFBC owing to the effect of the RBFNN.

From the above comparison, both control schemes successfully realize trajectory tracking control and achieve similar accuracy. However, differences exist. Unlike AFBC, ARBFBC is derived with inertial matrix and decoupling. Furthermore, the proposed AFBC achieves fixed-time stability; that is, all the trajectory tracking errors and adaptive gains converge at approximately 8 s in the three cases. For ARBFBC, the convergence time of error increases as initial error increases. Although the increase is insignificant, the Lyapunov function in ARBFBC comprises the error and adaptive gain. As the error increases, the adaptive gain of ARBFBC adjusts to a high value and remains stable for an extended period (Fig. 10). The large gain can accelerate the convergence of the error, which explains why the convergence time of the error does not increase apparently. Nevertheless, the Lyapunov function does not converge when the error converges, owing to the slow convergence of the gain. Consequently, the control input oscillates significantly. In contrast, the proposed method forces the error and coefficient to converge before 8.5 s, achieving the convergence of the Lyapunov function in a fixed time. The comparison results are summarized as follows.

- (1) The proposed AFBC scheme achieves fixed-time stability, inducing the convergence of the trajectory tracking error and adaptive gain within 8.5 s in all three cases. It generates a smooth and easily executable control input, whereas ARBFBC does not.
- (2) With the similar accuracy, faster convergence rate, and lower vibration, AFBC is derived without decoupling and model parameters, whereas ARBFBC is derived with an inertial matrix.
- (3) The proposed AFBC scheme tackles model uncertainty by a state-dependent adaptive gain, whereas ARBFBC addresses it with an NN and adaptive gain. Nevertheless, the gain of ARBFBC is smaller than that of AFBC after convergence, owing to the effect of the RBFNN. In the case of severe external disturbances, the adaptive gain in AFBC may increase significantly and cause chattering.

5. Conclusion

An adaptive fixed-time backstepping control scheme under model uncertainty and external disturbances is developed to achieve 3D trajectory tracking of underactuated AUVs. In this method, a novel virtual

velocity guidance law is derived using the backstepping method. Subsequently, an adaptive fixed-time velocity controller is derived without model parameters. Lyapunov stability analysis indicates that the proposed controller guarantees trajectory tracking error converging to approximately zero within a fixed time. Simulations of 3D trajectory tracking in the proposed control scheme and a comparison between the proposed method and ARBFBC are performed to verify the effectiveness and superiority of the proposed scheme.

CRedit authorship contribution statement

Hongxuan Chen: Methodology, Software, Writing – original draft, Writing – review & editing. **Guoyuan Tang:** Conceptualization, Ideas, Supervision, Funding acquisition. **Shufeng Wang:** Formal analysis, Validation. **Wenxuan Guo:** Investigation, Project administration. **Hui Huang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgment

This study is supported by the National Natural Science Foundation of China (Grant Number 51979116).

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